Pranav Reddy
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PROJECTS & EXPERIENCE

Amazon Web Services, Braket
SDE Intern (Full Time Return Offer) June 2026 — September 2026

- Built an optimizing quantum compiler lowering OpenQASM through HUGR IR to device native circuits, with qubit placement, routing, and native gate decomposition (Qiskit, tket).
- Extended it to dynamic circuits (mid circuit measurement, feed forward), with per device compile time validation of control flow constraints.

Anduril Industries Capstone: LWIR Metasurface Polarimeter
Photonics Software Lead September 2026 — Present

- Swept 80 wire grid metasurface polarizer geometries across 6–14 µm. The best design reached 64% transmission and a 3,750:1 extinction ratio over 8–12 µm, 4.7× and 5× the prior year design.
- Built a Python polarimetric imaging simulator (Mueller matrix pixels, micropolarizer mosaic, Stokes/DoLP) to quantify how extinction ratio and pixel misregistration degrade accuracy.

Optical Interconnect Codesign (OICD)
Independent Project

- Built an open source Python link budget engine for copackaged optics, validated against the draft IEEE 802.3dj budget.
- Showed 16 wavelength microring BiDi links need FSR ≥ 3.2 THz. Reimplemented a photonic tensor core's block floating point datapath (*Nature* 2025), cutting matrix vector error from 6.5% to 2.0%.

Quantum Codesign Lab
Niu Group, UCSB Undergraduate Researcher June 2025 — Present

- Developing a photonic interconnect simulator for distributed quantum computing (DARPA HARQ), propagating optical link loss and noise to system level error rates.
- With UCSD Picasso Lab (Prof. Yufei Ding), extending LightStim into a physical level compiler that lowers neutral atom error correction circuits into gate and atom movement schedules.
- Building a benchmark that evaluates LLM agents on cat qubit hardware design tasks.

UCSB Photonics & Optical Systems Labs
Lab Experimentalist & Coursework Researcher June 2024 — Present

- Simulated hollow core fiber and 2.5–40 Gb/s WDM links in VPIphotonics, analyzing modulator chirp, extinction ratio, dispersion, crosstalk, and BER vs. SMF/DCF over 80–200 km.
- Measured single photon antibunching ($g^2(0) \approx 0.13$) and a CHSH violation ($S = 2.35$) with a heralded SPDC source and SPAD coincidence counting.

EDUCATION

BS/MS ECE (+1 Program, Electronics & Photonics) | UC Santa Barbara Aug 2023 — May 2028

COURSE WORK

Fiber Optics	(ECE 135)	Optical Systems	(ECE 136A)	Quantum Photonics	(ECE 136C)
Waveguide Physics	(ECE 144)	Signal Processing	(ECE 130)	Optoelectronic Physics	(ECE 162)

SKILLS

Programming: Python, C++, C, MATLAB
Photonics: VPIphotonics, Strawberry Fields, transfer matrix/finite difference methods, link budgets, SPDC/SPAD setups
Quantum & Systems: Qiskit, tket, HUGR, OpenQASM, STM32, Git, Linux